

Machine Learning

FIB, Master in Data Science

Marta Arias, Computer Science @ UPC

Course information

Instructors

Marta Arias (theory + lab groups 11+12)

- ▶ marias@cs.upc.edu
- ▶ no office hours; if you want to talk, email me and we'll set a time

Ignasi Gomez (lab groups 13+14)

- ▶ ignasi.gomez@upc.edu

Class logistics

- ▶ Course material (lecture slides, lab notebooks, project info) placed weekly through the **racó**
- ▶ Announcements and submissions through the **racó**
- ▶ Theory lectures (Mondays 4-6pm)
 - ▶ mostly presentations with slides, with occasional “old-school” chalk-board class
 - ▶ lecture notes/slides posted as we progress through the course
- ▶ Lab sessions (Tuesdays 2-4pm and 4-6pm)
 - ▶ will use **python notebooks** mostly
 - ▶ notebooks will implement and illustrate concepts from theory and introduce libraries as well
 - ▶ maybe use labs to do some examples in depth
 - ▶ may propose exercises and examples to solve *in your own time*, not graded
 - ▶ may introduce new complementary topics with examples to what is presented in theory classes

Evaluation

- ▶ $\text{Grade} = 20\% \text{ mid-term} + 40\% \text{ exam} + 40\% \text{ project}$

Course project

- ▶ Done in **pairs** within same lab group (this is *strict*)
- ▶ Dataset and rules TBA; probably presented as a private Kaggle competition
- ▶ Evaluation mainly via a final **written report** (along with the code)

Delivery of project report and code towards the **end of June**; exact date will be announced in the coming weeks

Mission statement

*The goal of this course is to introduce fundamental **concepts** in machine learning and key methodologies. Rather than focusing on the latest developments—which emerge rapidly—it aims to provide a strong foundation, enabling you to understand and engage with new advancements in the field.*

Contents¹

Supervised learning:

- ▶ Linear methods:
 - ▶ Linear methods for regression
 - ▶ Linear methods for classification
- ▶ Non-linear methods:
 - ▶ Kernel methods (support vector machines)
 - ▶ Artificial Neural Networks
 - ▶ Random Forests and other ensemble methods

Unsupervised learning:

- ▶ Dimensionality reduction
- ▶ Clustering

¹Disclaimer: the topics and/or their order may change (slightly)

Main bibliography

- ▶ **Pattern Recognition and Machine Learning**
Christopher M. Bishop, Springer, 2006
- ▶ **The Elements of Statistical Learning**
Hastie, Tibshirani and Friedman (2009). Springer-Verlag.
- ▶ **Machine Learning: a Probabilistic Perspective**
Kevin P. Murphy, MIT Press 2012; new edition [drafts here](#)
- ▶ **Introduction to Machine Learning**
Ethem Alpaydin (3rd Ed.), The MIT Press, 2015
- ▶ **Mathematics for Machine Learning**
Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, Cambridge University Press, 2020.
- ▶ ... There's a whole web out there